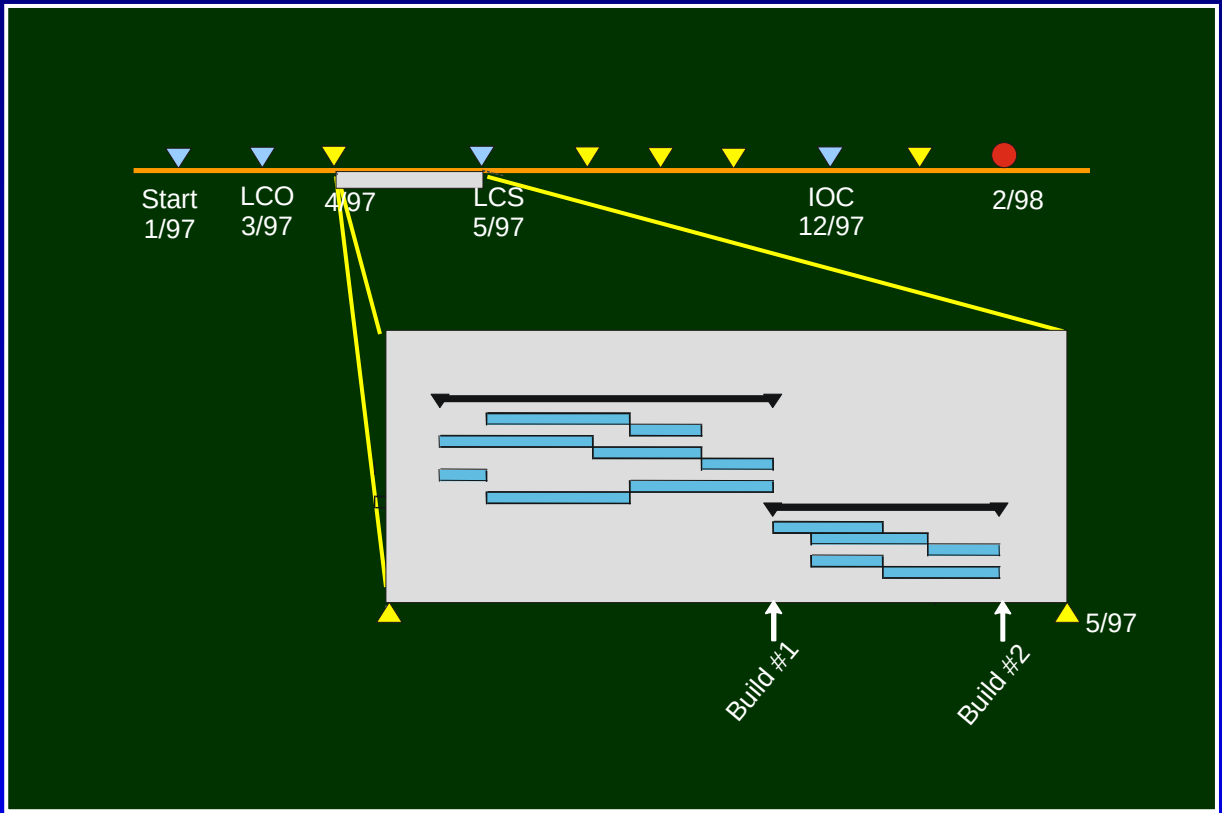
Project Management



Copyright © 1998 Rational Software, all rights reserved

R

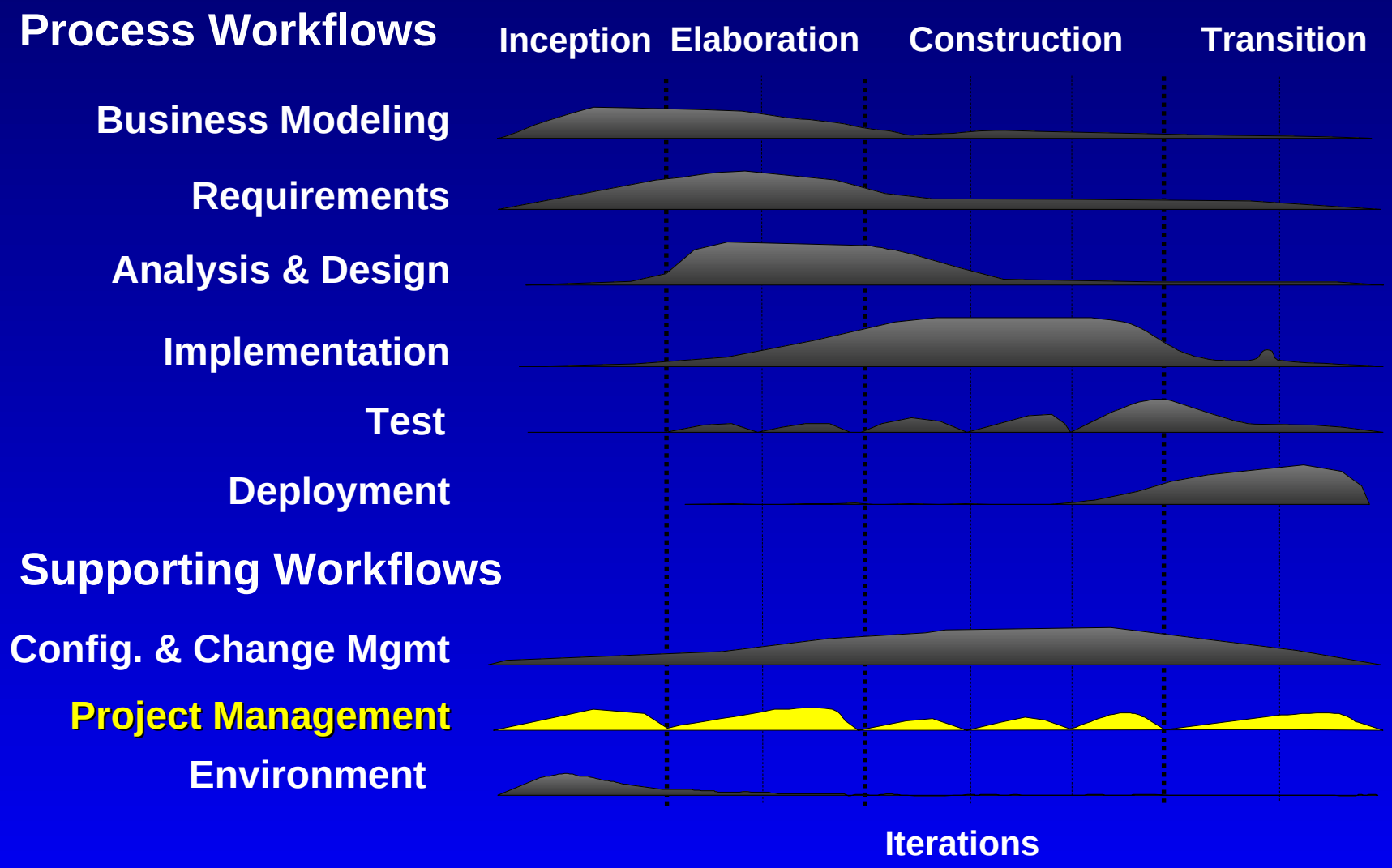
Objectives: Project Management

- Explain that process phases represent business decision points
- Understand the major and minor **milestones** of the process
- Explain how the iterative lifecycle is controlled through iteration **assessment**
- Explain the incremental approach to **project planning**
- Understand important **metrics** to measure project status and product quality
- Understand the management artifacts of the process

Copyright © 1998 Rational Software, all rights reserved

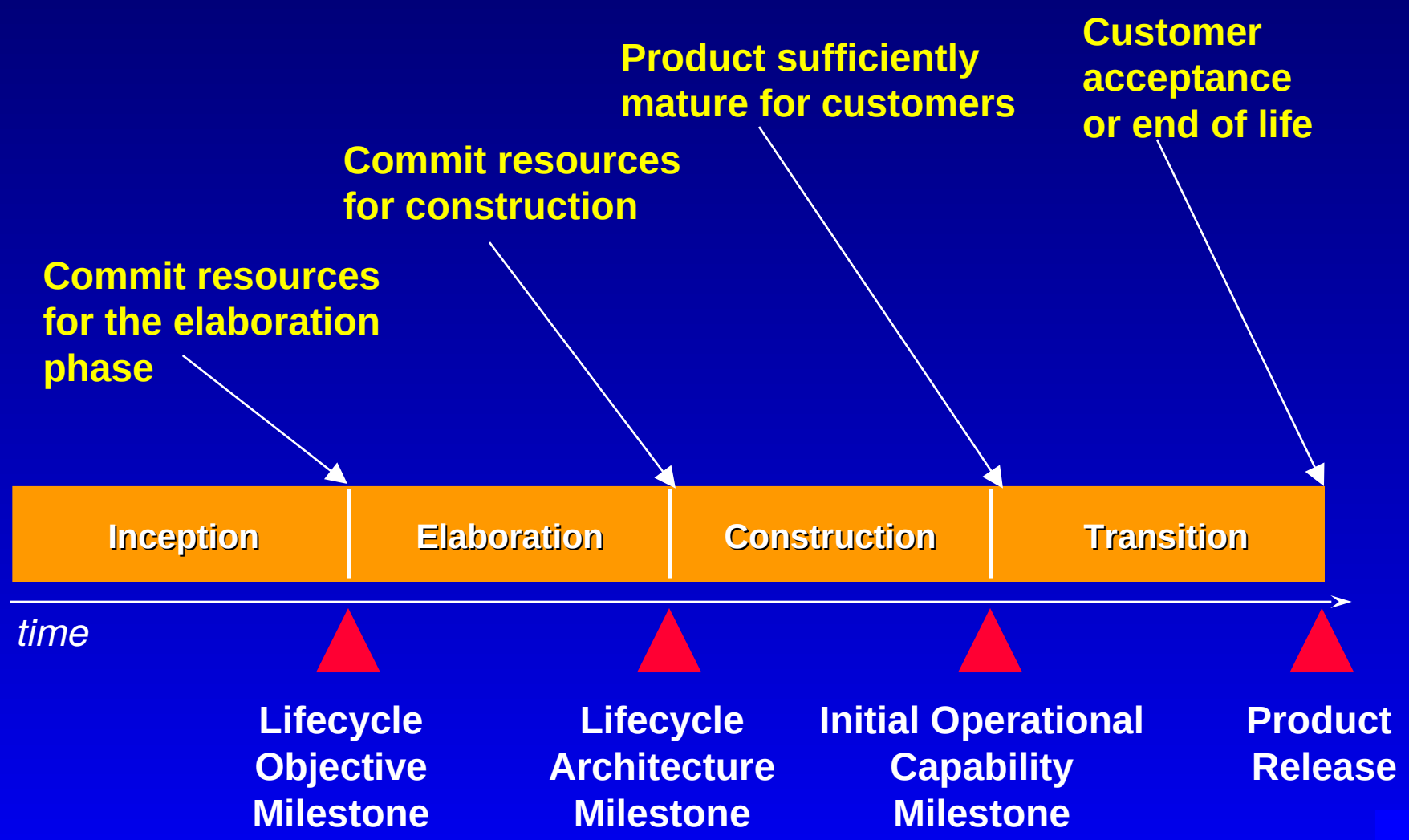
R

Project Management Has Its Own Workflow



Copyright © 1998 Rational Software, all rights reserved

Major Milestones: Business Decision Points



Copyright © 1998 Rational Software, all rights reserved

Inception Phase

- Evaluation criteria:
  - Stakeholder concurrence on scope definition and cost/schedule estimates
  - Requirements understanding for the primary use cases
  - Credibility of the estimates, priorities, risks, development process, etc.
  - Depth and breadth of architectural prototype (optional)
  - Actual expenditures versus planned expenditures
- Milestone: Lifecycle Objectives (LCO)

Copyright © 1998 Rational Software, all rights reserved

Elaboration Phase

- Evaluation criteria:
  - Vision of the product is stable
  - Requirements understanding (use-case model >= 80% complete and supplementary specifications complete)
  - Executable architecture baselined and stable
  - Major technical risks ‘retired’
  - Sufficient software development plan for construction
  - Stakeholder agreement that the vision can be met with the plan and architecture in place
  - Actual expenditures versus planned expenditures
- Milestone: Lifecycle Architecture (LCA)

Copyright © 1998 Rational Software, all rights reserved

Construction Phase

- Evaluation criteria
  - Complete product of sufficient quality available
  - Deployment plan ready
    - Packaging
    - Pricing
    - Roll-out
    - Support
    - Training
    - Production
    - Transition strategy
  - User documentation available
- Milestone: Initial Operational Capability (IOC) "beta"

Copyright © 1998 Rational Software, all rights reserved

R

Transition Phase

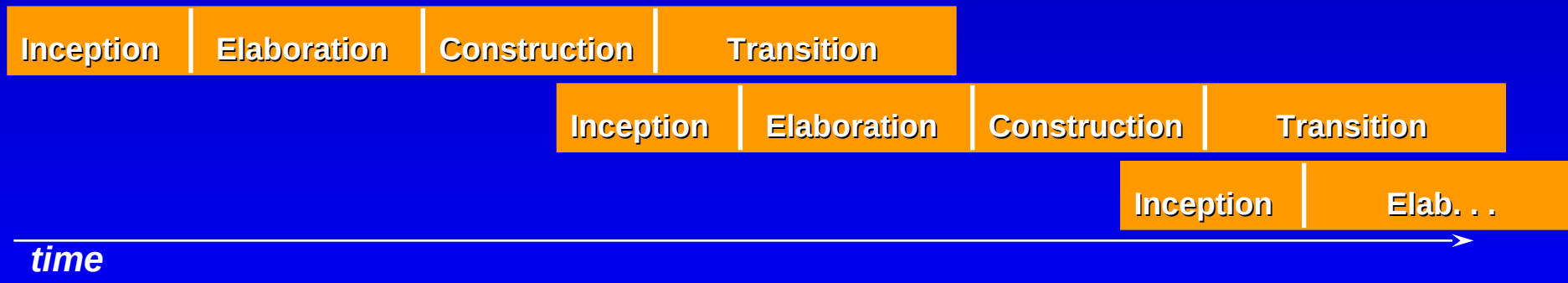
- Evaluation criteria
  - Previous products and artifacts, updated as necessary
  - Customer/user acceptance of released product
  - Post-mortem analysis of the organization's performance; additional assets; potential evolutions
- Milestone: Product Release ("GA" or general availability release)

Copyright © 1998 Rational Software, all rights reserved

R

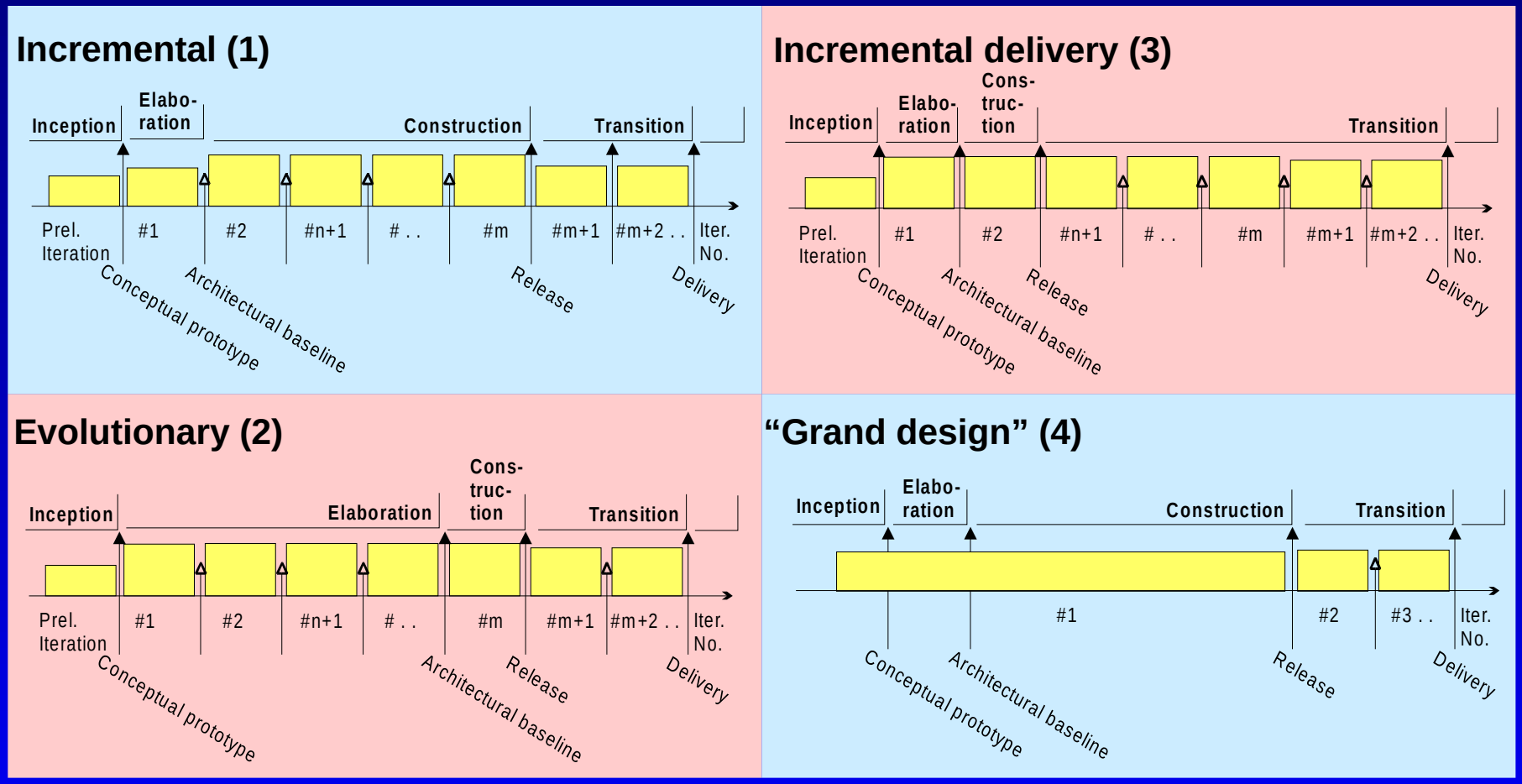
Beyond Transition: Development Cycles

- A **development cycle** includes one execution of all four phases and produces a software **generation**
- Most software systems require multiple cycles
  - An initial development cycle leading to the initial delivery
  - Subsequent cycles to maintain and enhance the system
- Cycles may be sequential, but more commonly overlap



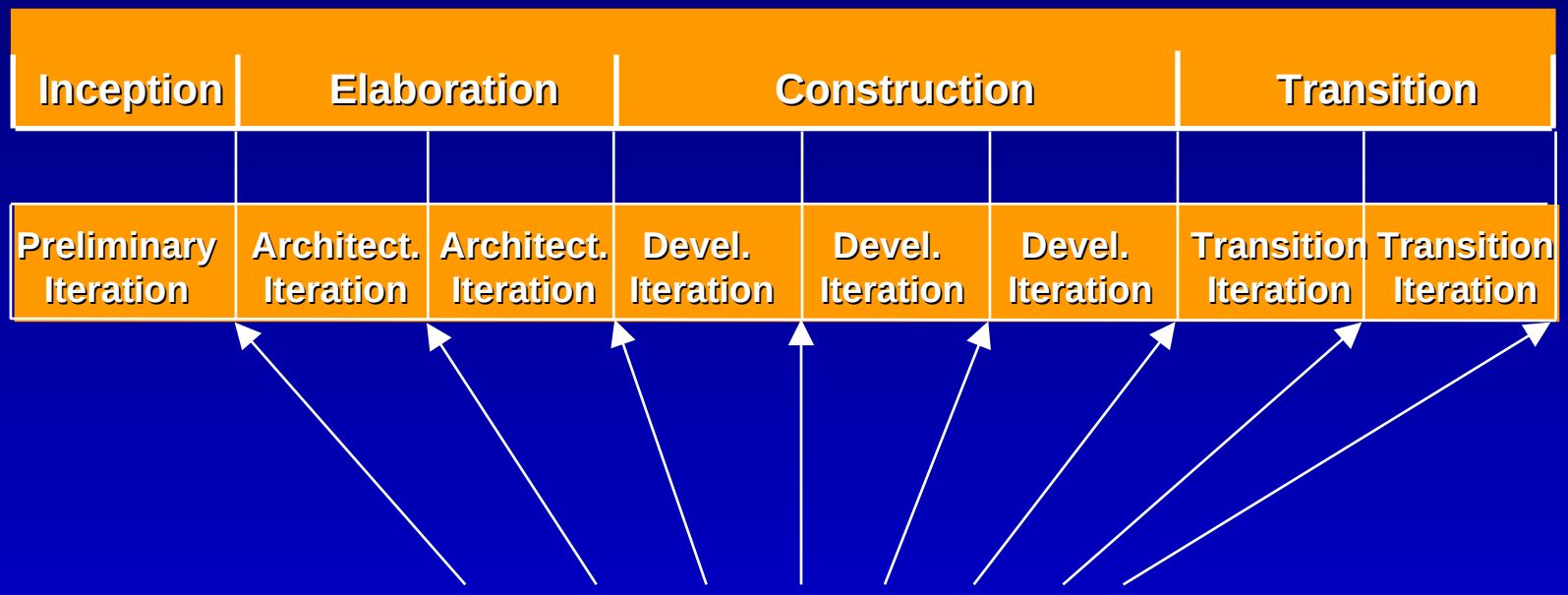
Copyright © 1998 Rational Software, all rights reserved

Strategies for Iterative Development



Copyright © 1998 Rational Software, all rights reserved

Minor Milestones: Technical Visibility Points



Minor Milestones: Releases

Copyright © 1998 Rational Software, all rights reserved

R

Minor Milestones: Iteration Feedback Loop

Iteration N Cost and Schedule Actuals

Quality Assessment for Iteration N

- Test Results
- Defect Density
- Architecture Stability
- Other metrics

Iteration N Assessment

- Compare iteration actual cost, schedule, and content with iteration plan
- Determine rework (if any) to be done
  - Assign to future iteration(s)
- Determine what risks have been eliminated, reduced, or newly identified in this iteration
- Update project plan
- Prepare iteration plan for next iteration
  - Use revised risk list and select appropriate scenarios

Revised Risk List

Revised Project Plan

- Total Cost
- Overall Schedule
- Scope/Content

Iteration N+1 Plan

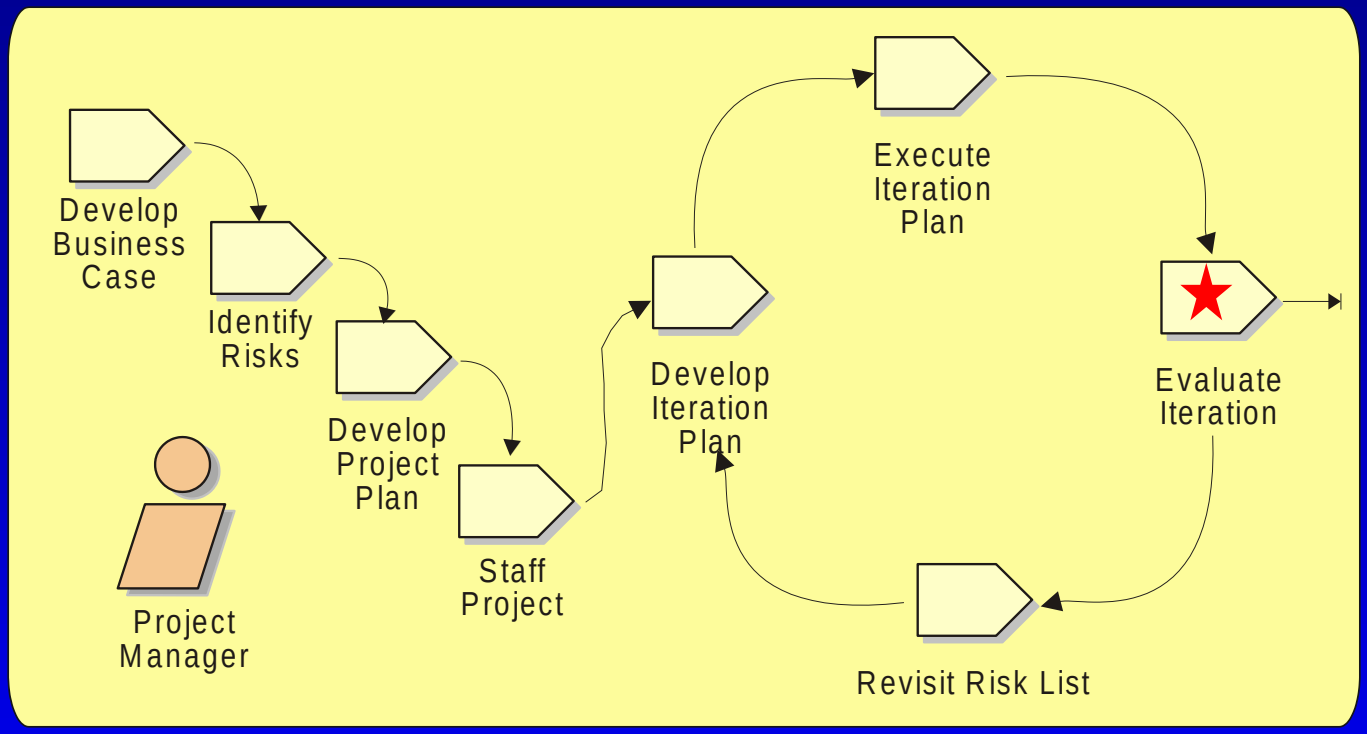
- Cost
- Schedule
- Content

R

Copyright © 1998 Rational Software, all rights reserved

Controlling the Iterative Lifecycle

- Iteration assessments provide the feedback loop to allow us to control the process



Copyright © 1998 Rational Software, all rights reserved

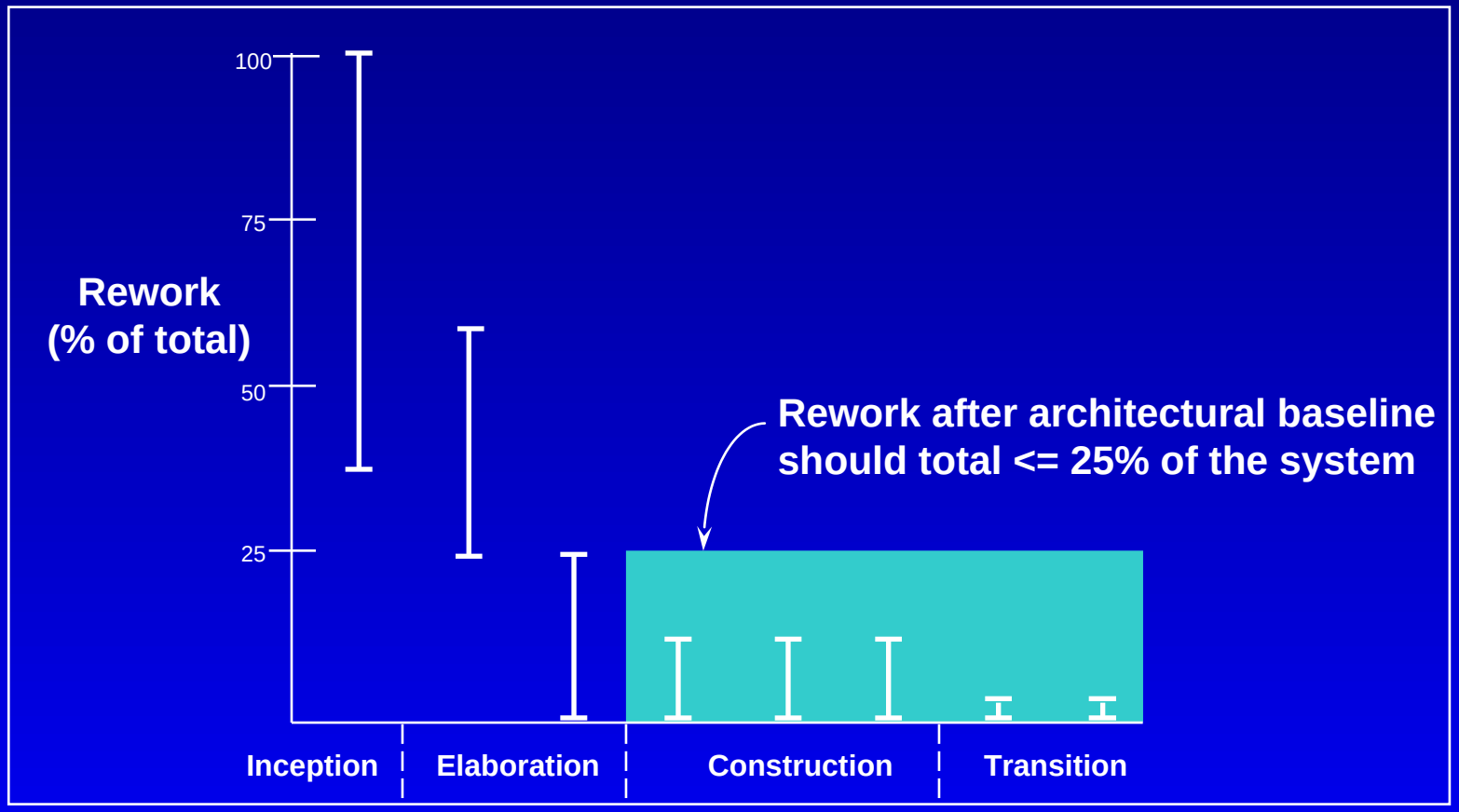
What Does a Project Need to Measure?

Metric	Meaning
Progress	Size and complexity
Stability	Rate of change in the project's complexity or size
Adaptability	Ease of change
Modularity	Scope of change
Quality	Number of errors
Maturity	Frequency of errors
Expenditures	Project expenditures against plan

Copyright © 1998 Rational Software, all rights reserved

Sample Stability Metric

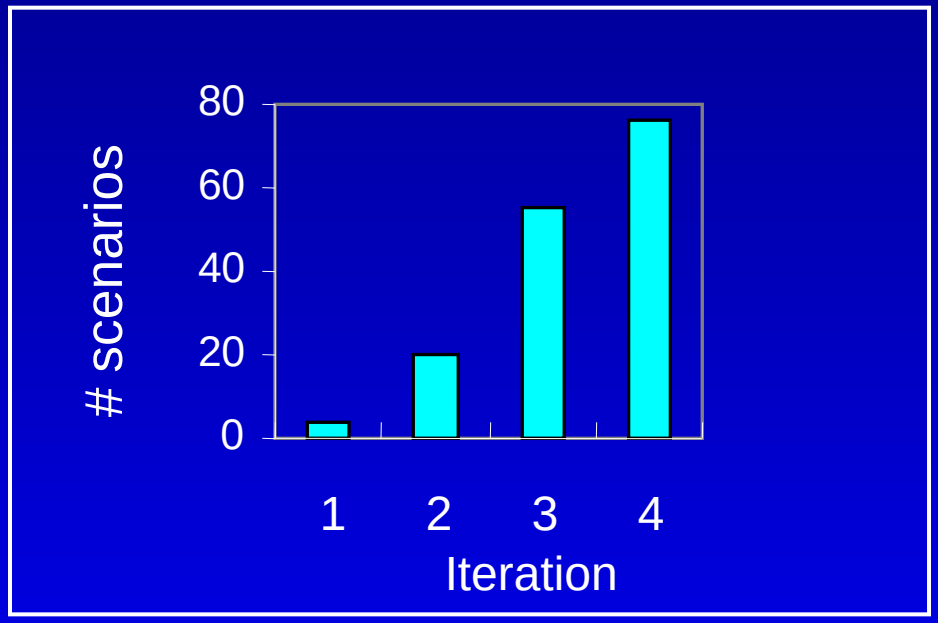
Rework Trends



Copyright © 1998 Rational Software, all rights reserved

Sample Progress Metric

Scenario Completion by Iteration

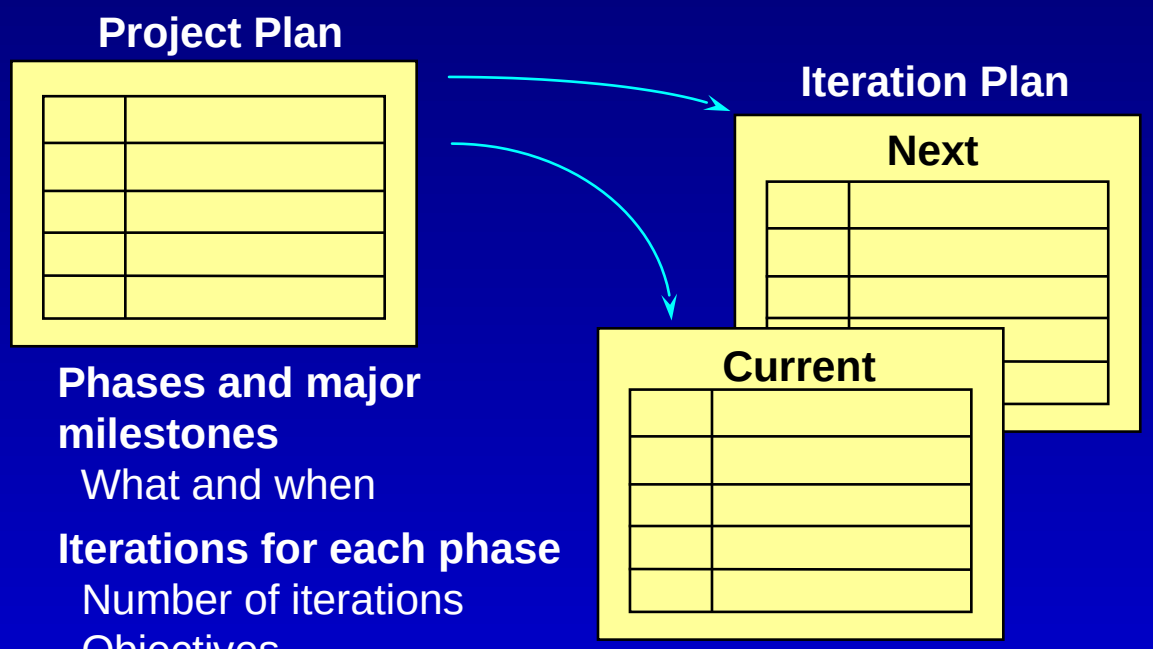


Data from a large (750 KSLOC) project:

- 6-month iterations
- 3 elaboration iterations
- 4 construction iterations (one completed)
- 2 transition iterations planned

Copyright © 1998 Rational Software, all rights reserved

Planning Is Incremental



Phases and major milestones
- What and when

Iterations for each phase
- Number of iterations
- Objectives
- Duration

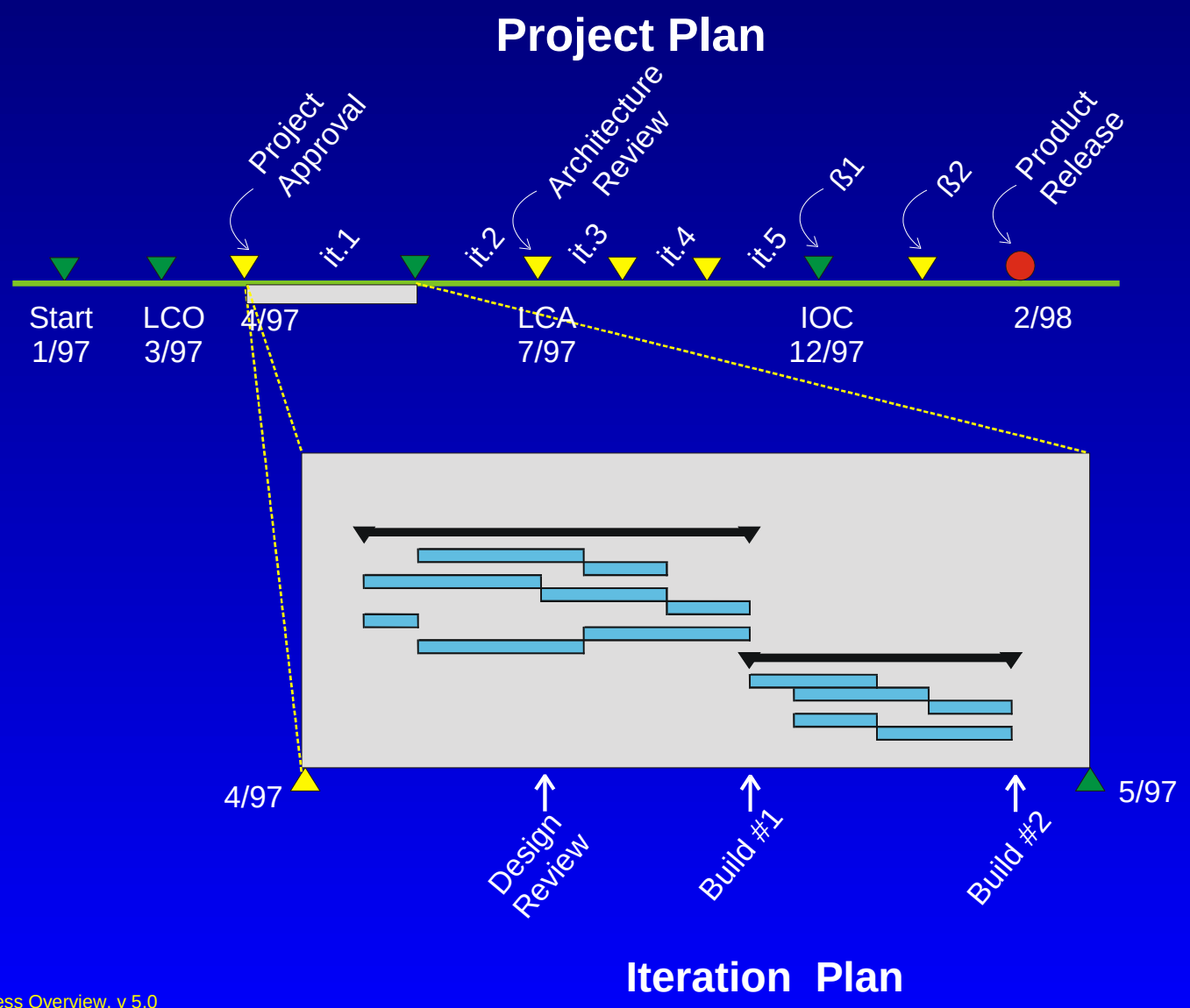
Staffing profile

Tasks
Allocation to individuals/teams
Major builds

Coarse-grained Plan **Fine-grained Plans**

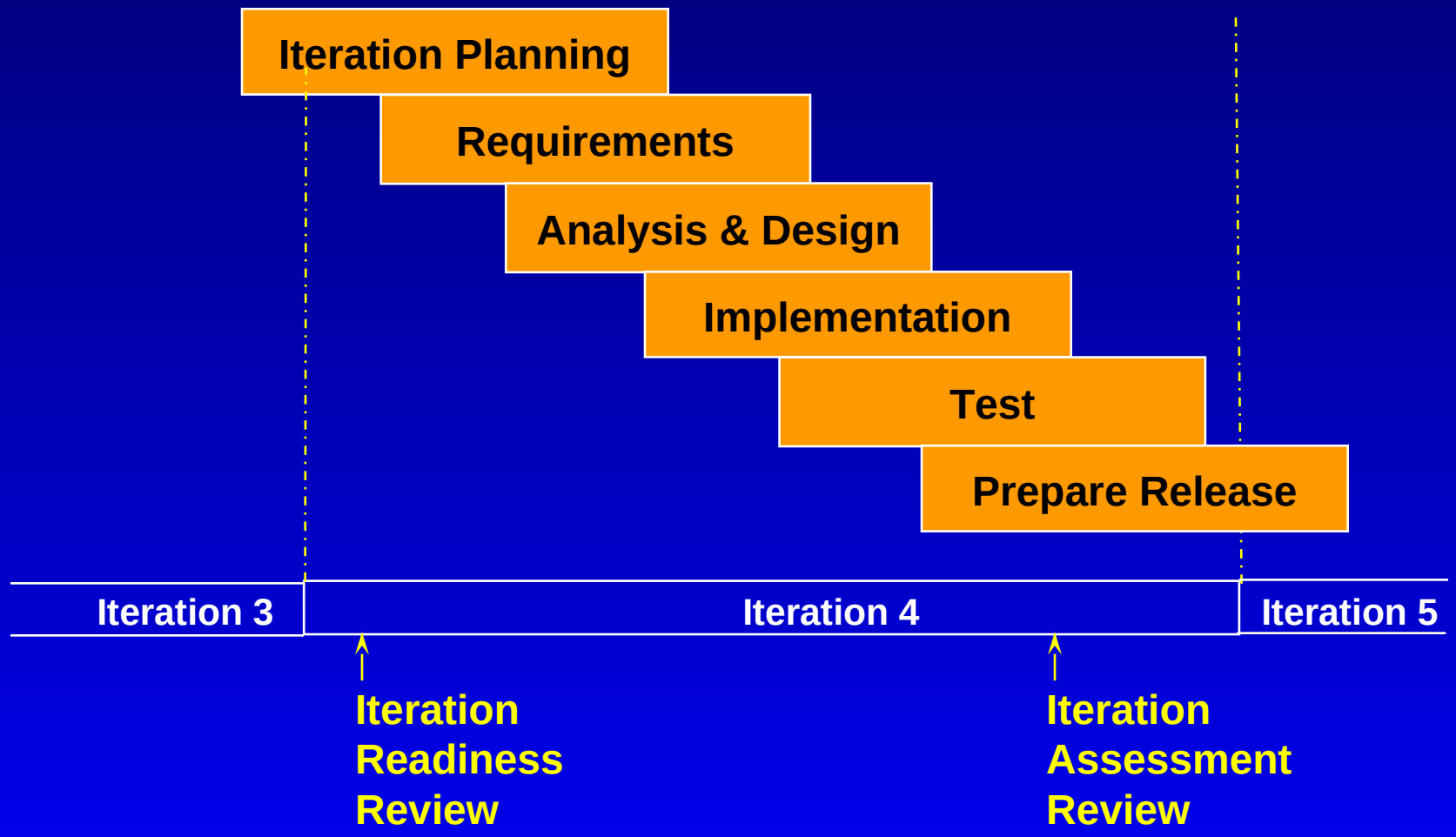
Copyright © 1998 Rational Software, all rights reserved

Project Plan and Iteration Plan



Copyright © 1998 Rational Software, all rights reserved

A Single Iteration Contains a "Mini-Waterfall"



Copyright © 1998 Rational Software, all rights reserved

Iteration Planning

- Define objective evaluation criteria
- Identify what concrete, measurable artifacts will be developed or updated and the activities required to build them
- Use a standard work breakdown structure
- Use estimates to assign duration and effort to each activity
- Adjust as necessary to keep all numbers within resource constraints

Copyright © 1998 Rational Software, all rights reserved

How Many and How Long?

- How many iterations should be included in the project plan?

	Total	I	E	C	T
Low	**3**	**0**	**1**	**1**	**1**
Typical	**6**	**1**	**2**	**2**	**1**
High	**9**	**1**	**3**	**3**	**2**

- How long should each iteration take?
  - Depends on a number of factors:
    - Size of the system being built: The larger the system, the longer the duration
    - Number of people: The larger the number of people, the longer the duration

Copyright © 1998 Rational Software, all rights reserved

Risk Terms

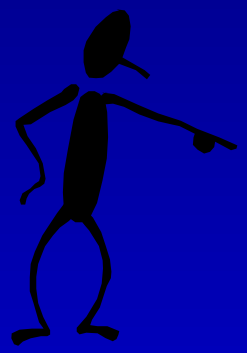
- **Risk** - whatever may stand in the way of our success
- **Direct risk** - the project has a large degree of control
- **Indirect risk** - the project has little or no control
- Risk attributes:
  - Probability of occurrence
  - Impact on the project (severity)
- Risk magnitude indicator:
  - High, Significant, Moderate, Minor, Low

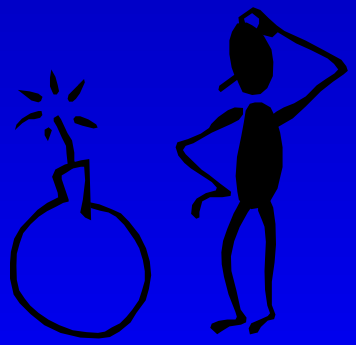
Copyright © 1998 Rational Software, all rights reserved

R

Risk Strategies

- Risk avoidance - reorganize the project so that it will not be affected by the risk
- Risk transfer - reorganize the project so that someone else bears the risk
- Risk acceptance - live with it
  - Risk mitigation - reduce the probability or impact
  - Definition of a contingency plan - what course of action to take if the risk becomes an actual problem ("Plan B")





R

Copyright © 1998 Rational Software, all rights reserved

Some Sample Risks

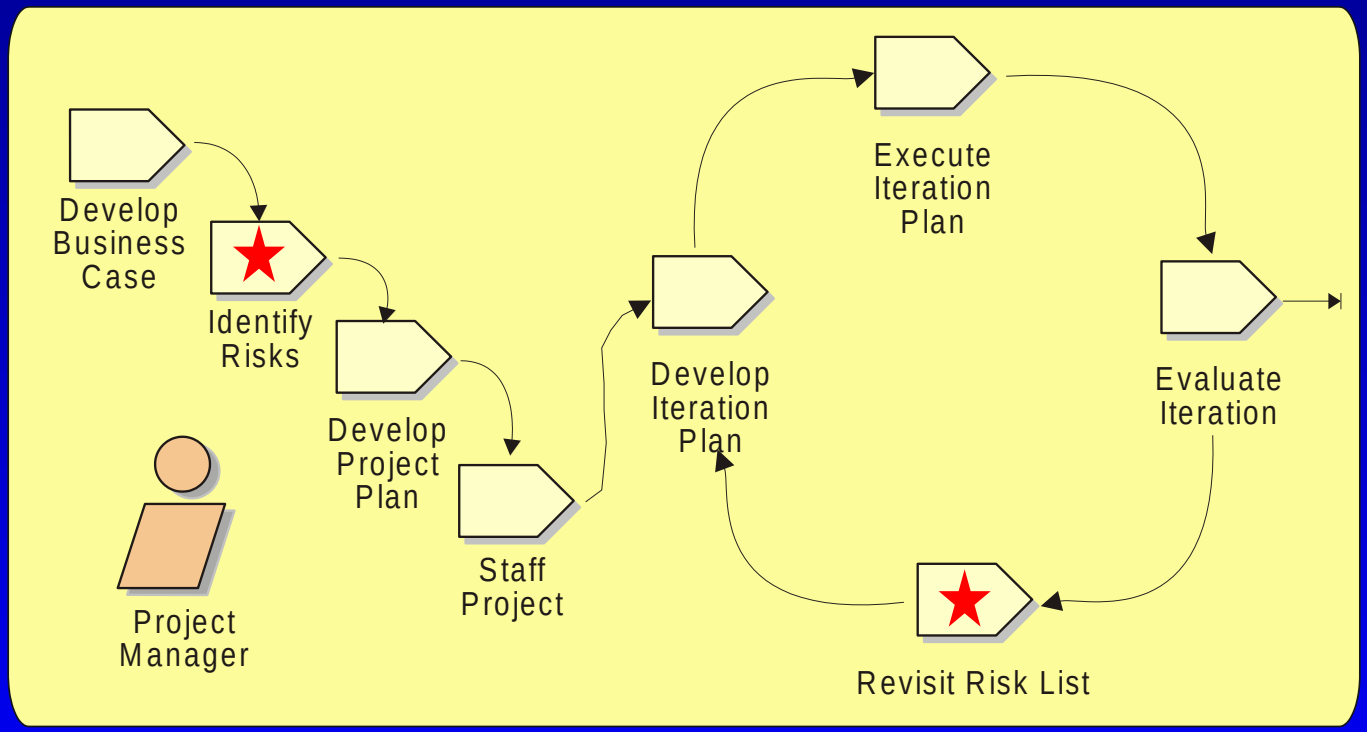
- Resource risks
  - People, skills, funding,. . .
- Business risks
  - Competition, ROI, supplier interfaces,. . .
- Technical risks
  - Unproven technology, uncertain scope,. . .
- Schedule risks
  - Only 24 hours in a day,. . .

Copyright © 1998 Rational Software, all rights reserved

R

Risks Drive the Iterative Lifecycle

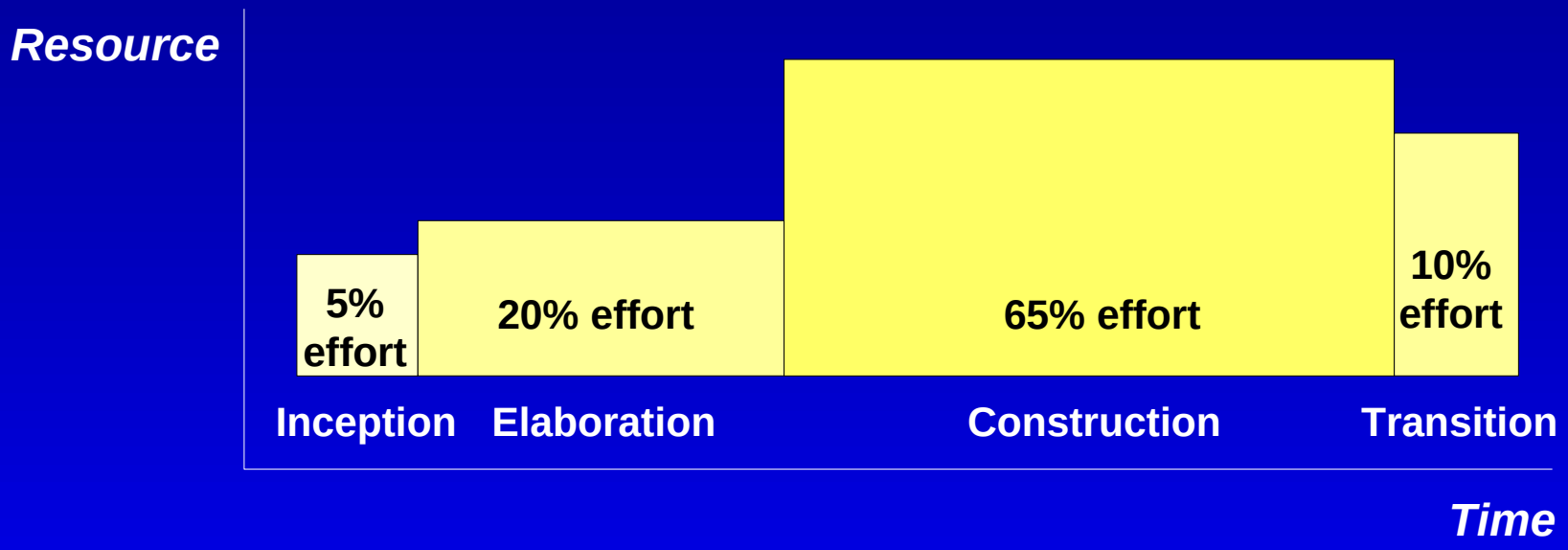
- Risk assessment is a continuous process; risks change over time
- Early iterations address the highest risks



Copyright © 1998 Rational Software, all rights reserved

Staffing: Schedule and Effort

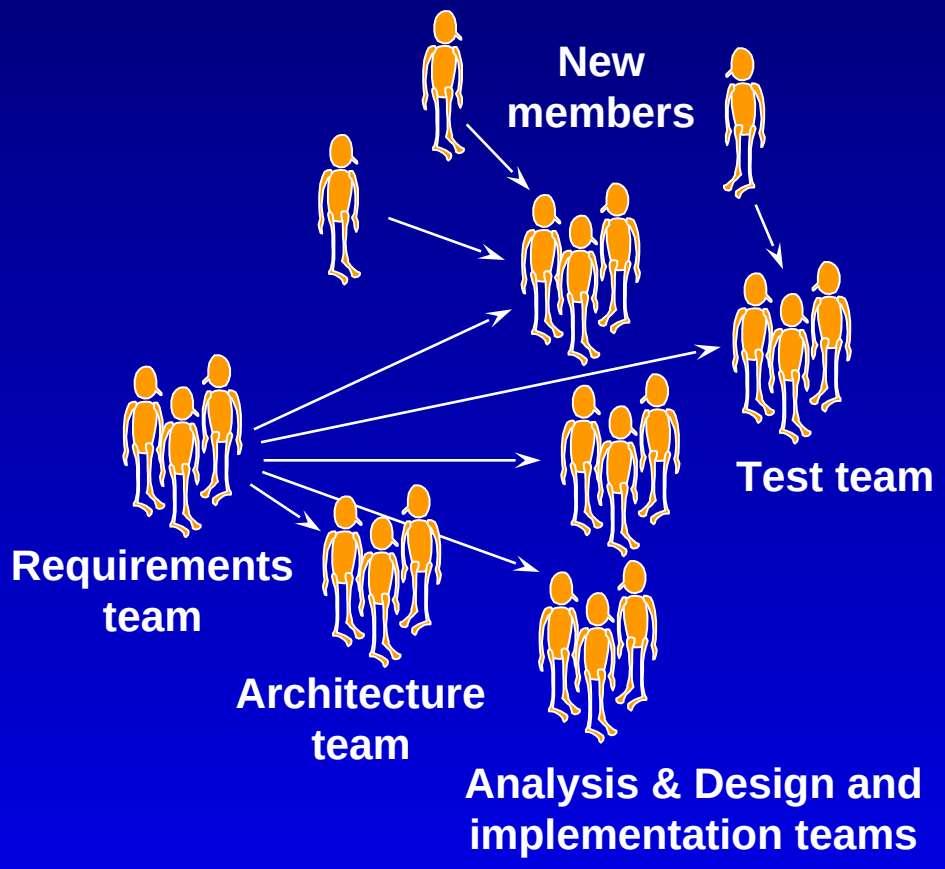
The anticipated distribution for a typical initial development cycle for a medium-sized project



Copyright © 1998 Rational Software, all rights reserved

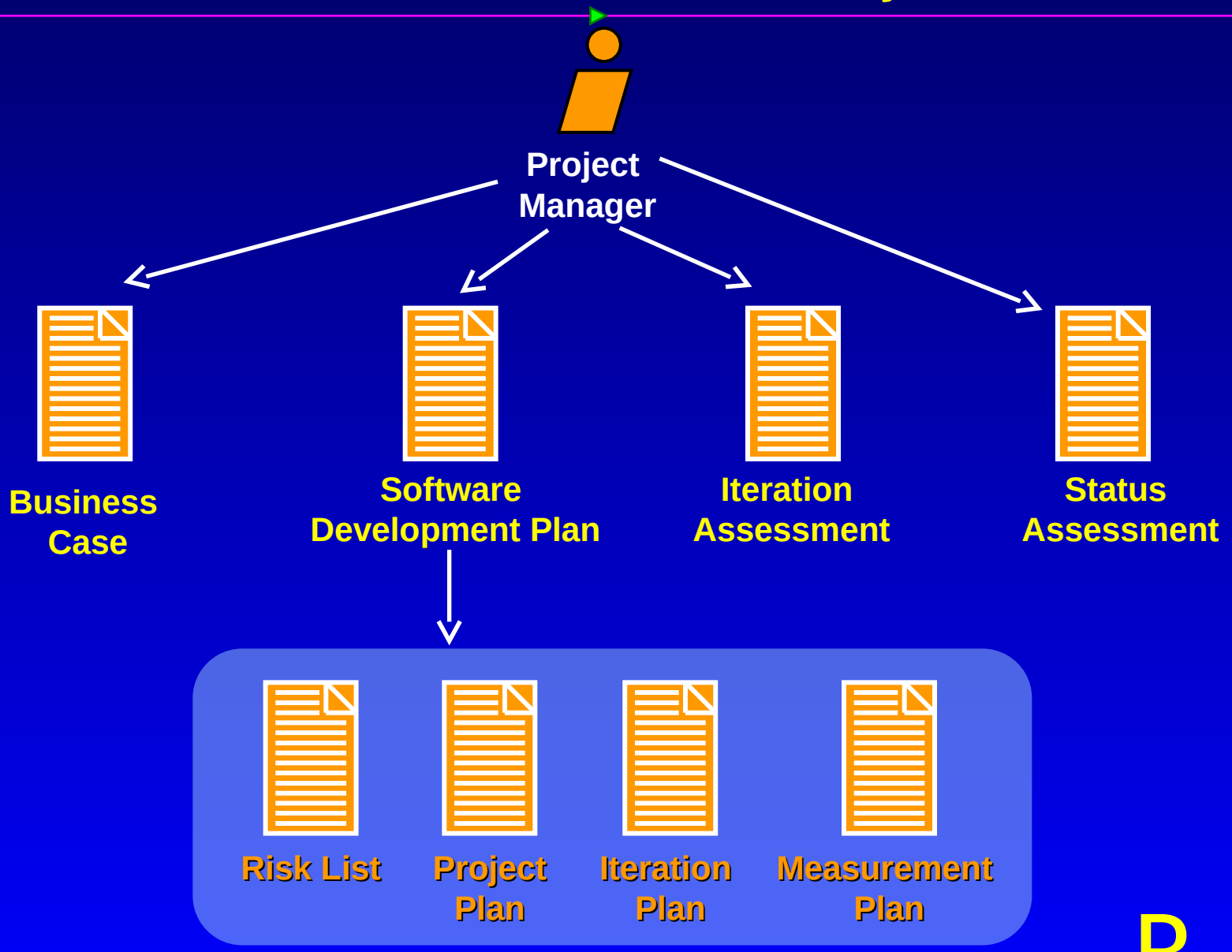
Project Organization: Teams

- Use the same people in requirements through implementation
  - Minimize hand-over
- Architecture team
- Well-balanced analysis & design teams
- A separate test team
- In general, teams should not be larger than 5-7 people



Copyright © 1998 Rational Software, all rights reserved

Artifacts to Plan and Control the Project



Copyright © 1998 Rational Software, all rights reserved

Summary: Project Management

- Phase boundaries represent major decision points
- Iteration boundaries represent technical visibility points
- Planning proceeds incrementally
  - overall project plan >> iteration plans
- Metrics are collected and evaluated at the end of each iteration
- Iteration assessments provide the feedback loop by which the iterative life cycle is controlled

Copyright © 1998 Rational Software, all rights reserved

R